

Database ERD and Normalization Report

Anonymous Group Submission

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1 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram below models the database architecture required to track how socioeconomic deprivation metrics correlate with hospital admissions. The diagram uses Crow's foot notation to illustrate the cardinalities between our core entities.

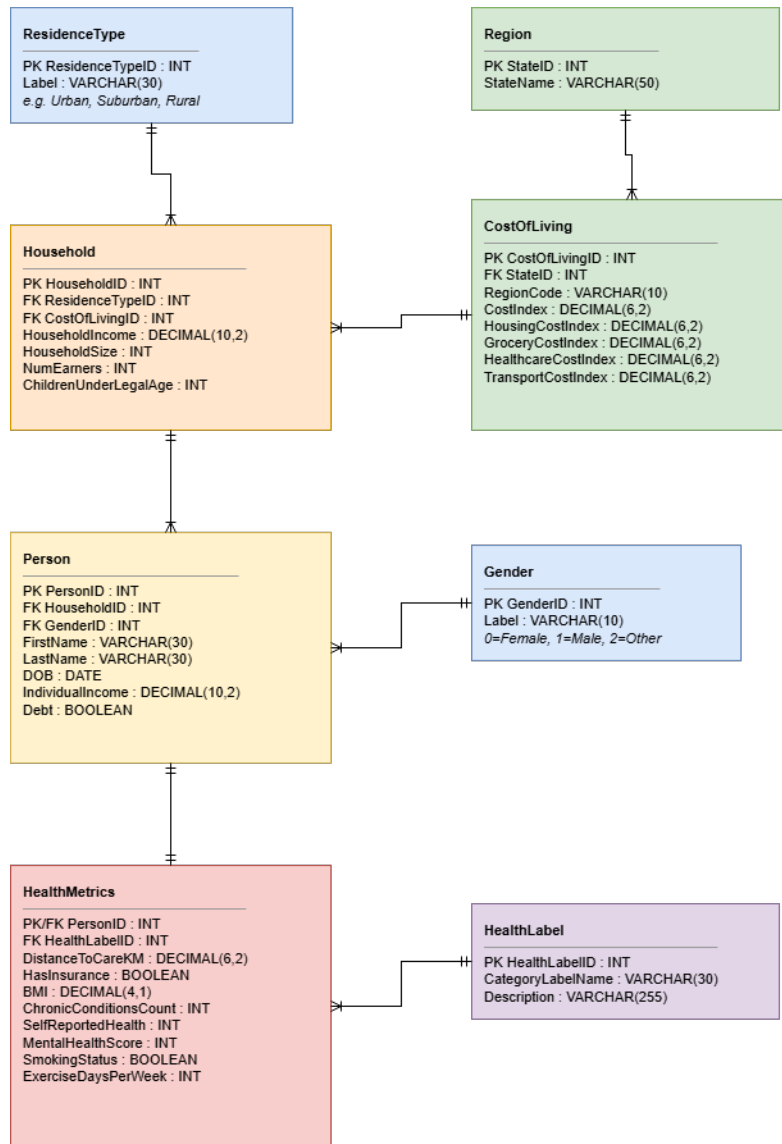


Figure 1: Final ERD in 3rd Normal Form (3NF).

2 Database Normalization Process

To ensure data integrity, eliminate redundancies, and avoid update anomalies, our database schema was progressively normalized from a raw dataset up to the Third Normal Form (3NF).

2.1 First Normal Form (1NF): Atomicity

Rule: Every cell must contain exactly one single, indivisible (atomic) value, and there must be no repeating groups.

Violation (Before 1NF): The initial dataset recorded demographic data at the household level, squeezing multiple people into a single column.

HouseholdID	Residence_Label	Household_Members (FirstName, DOB, Gender)
101	Urban - Apartment	John, 1980-05-12, Male — Jane, 1982-11-23, Female
102	Rural - House	Mark, 1995-02-14, Male

Table 1: Unnormalized Table (Violation of 1NF)

The Fix (After 1NF): We split the multi-valued attributes so each individual gets their own row, forming the baseline for our **Person** entity containing atomic attributes like **FirstName** and **DOB** (see Figure 1).

2.2 Second Normal Form (2NF): Eliminating Partial Dependencies

Rule: The table must be in 1NF, and all non-key columns must depend on the *entire* primary key (relevant for composite keys).

Violation (Before 2NF): If we use a composite key of (**HouseholdID**, **PersonID**), household-level attributes like **HouseholdIncome**, **HouseholdSize**, and **NumEarners** only depend on the **HouseholdID**, not the specific person (see Figure 1). This causes massive data duplication across family members.

HouseholdID (PK)	PersonID (PK)	FirstName	HouseholdIncome	NumEarners
101	P-01	John	85000.00	2
101	P-02	Jane	85000.00	2
102	P-03	Mark	42000.00	1

Table 2: Table in 1NF (Violation of 2NF)

The Fix (After 2NF): We separated the data into two distinct tables: **Household** and **Person** (see Figure 1). The **Person** table now uses **HouseholdID** strictly as a Foreign Key to link back to the parent household.

2.3 Third Normal Form (3NF): Eliminating Transitive Dependencies

Rule: The table must be in 2NF, and non-key columns must not depend on other non-key columns.

Violation (Before 3NF): In our 2NF tables, we stored descriptive text directly in the main tables. For example, the **StateName** depends on the region, not directly on the **Household**, creating a transitive dependency (see Figure 1).

HouseholdID (PK)	CostOfLivingID	CostIndex	StateID	StateName
101	CL-01	115.5	S-01	California
102	CL-01	115.5	S-01	California

Table 3: Table in 2NF (Violation of 3NF)

The Fix (After 3NF): We extracted categorical and geographic data into separate reference tables (see Figure 1):

- **Regional Data:** We split geographic metrics into **Cost Of Living of region** (tracking **HousingCostIndex**, **HealthcareCostIndex**, etc.) and a dedicated **RegionID** table tracking the **StateName**.
- **Demographic Categories:** We moved text labels for **ResidenceType** (e.g., Urban/Suburban/Rural) and **Gender** (0=Female, 1=Male, 2=Other) into their own tables linked via Foreign Keys to prevent spelling anomalies and redundant text entries.